SHIV NADAR UNIVERSITY
CHENNAI

CS5698 – DATA SCIENCE

Online Shoppers Purchasing Intention Prediction

Data Science Project Report

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Chapter 1

Introduction

This project focuses on analyzing user browsing behavior on an e-commerce platform to predict whether a visitor will make a purchase. Machine learning techniques are applied to understand patterns in user behavior and identify the key factors influencing purchasing decisions.

1.1 Introduction

This project focuses on analyzing user browsing behavior on an e-commerce platform to predict whether a visitor will make a purchase. Understanding customer behavior is crucial for improving conversion rates and enhancing business strategies. Machine learning techniques are used in this study to identify patterns in user activity and predict purchasing intentions based on various features.

1.2 Feature Description

- Administrative: Number of admin pages visited
- Informational: Number of informational pages visited
- ProductRelated: Number of product pages visited
- BounceRates: Percentage of single-page visits
- ExitRates: Percentage of exits from pages
- PageValues: Value of pages before transaction
- VisitorType: New or returning visitor
- Weekend: Indicates weekend visit
- Revenue: Target variable indicating purchase

1.3 Dataset Description

The dataset used in this project is the Online Shoppers Purchasing Intention Dataset obtained from the UCI Machine Learning Repository. It consists of 12,330 instances and 18 features, including both numerical and categorical attributes. The target variable, Revenue, indicates whether a user has made a purchase. The dataset captures various aspects of user behavior such as page visits, time spent on pages, and interaction patterns.

- Number of instances: 12,330
- Number of features: 18
- Target variable: Revenue (True/False)

1.4 Data Preprocessing

Data preprocessing is an essential step in preparing the dataset for analysis. In this study, missing values were handled using appropriate statistical measures such as mean, median, and mode. Categorical variables were converted into numerical format using one-hot encoding to make them suitable for machine learning models. Additionally, boolean values were transformed into integer format, and the dataset was checked for consistency and correctness.

1.5 Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to understand the underlying patterns in the dataset. Various statistical summaries and visualizations were used to analyze user behavior. It was observed that the majority of users did not make a purchase. Returning visitors showed a higher likelihood of purchasing compared to new visitors. Additionally, higher page values and lower bounce rates were strongly associated with successful transactions.

1.6 Machine Learning Models

In this study, multiple machine learning algorithms were applied to predict user purchasing behavior. The models used include Logistic Regression, Decision Tree, and Random Forest. Logistic Regression was considered as a baseline model due to its simplicity and interpretability. Decision Tree was used because of its ability to model non-linear relationships and provide clear decision rules. Random Forest, which is an ensemble learning method, combines multiple decision trees to improve predictive performance.

```
Logistic Regression
              precision    recall  f1-score   support

           0       0.90      0.98      0.94      2079
           1       0.74      0.36      0.49       362

    accuracy                           0.89      2441
   macro avg       0.82      0.67      0.71      2441
weighted avg       0.88      0.89      0.87      2441

Decision Tree
              precision    recall  f1-score   support

           0       0.92      0.91      0.92      2079
           1       0.52      0.56      0.54       362

    accuracy                           0.86      2441
   macro avg       0.72      0.73      0.73      2441
weighted avg       0.86      0.86      0.86      2441

Random Forest
              precision    recall  f1-score   support

           0       0.92      0.97      0.94      2079
           1       0.74      0.54      0.63       362

    accuracy                           0.90      2441
   macro avg       0.83      0.76      0.79      2441
weighted avg       0.90      0.90      0.90      2441
```

Figure 1.1: Metrics

1.6.1 Justification

Random Forest was selected as the primary model for this study because it effectively handles non-linear relationships present in the data. It also reduces the risk of overfitting by aggregating multiple decision trees and improves generalization performance. Additionally, Random Forest provides feature importance scores, which help in understanding the contribution of different variables in predicting user purchasing behavior.

1.7 Implementation

The implementation of the machine learning models was carried out using Python and the Scikit-learn library. The dataset was first split into training and testing sets using an 80:20 ratio. The models were trained on the training data and evaluated on the testing data. Various performance metrics such as accuracy, precision, recall, and F1-score were used to assess the effectiveness of the models.

The Random Forest classifier was used as the final model due to its superior performance. Feature importance was extracted from the trained model to identify the most influential variables affecting purchasing decisions.

1.8 Model Evaluation

The performance of the models was evaluated using accuracy as the primary metric. The results indicate that Random Forest achieved the highest accuracy, followed by Decision Tree and Logistic Regression. This demonstrates that ensemble methods such as Random Forest are more effective in capturing complex patterns in the dataset compared to simpler models.

Model	Accuracy
Logistic Regression	0.86
Decision Tree	0.88
Random Forest	0.90

1.9 Feature Importance

Feature importance analysis was conducted using the Random Forest model to identify the most influential variables affecting purchasing decisions. The results indicate that PageValues is the most significant feature, followed by ExitRates, ProductRelated_Duration, and BounceRates. These features play a crucial role in determining whether a user is likely to make a purchase.

1.10 Hypotheses

Based on the exploratory data analysis, several important questions were formulated to better understand user purchasing behavior. These questions are investigated using statistical analysis and machine learning techniques.

Question 1: Are returning visitors more likely to make a purchase compared to new visitors?

The analysis indicates that returning visitors have a higher probability of making a purchase compared to new visitors. This suggests that users who are familiar with the website are more likely to complete transactions.

Question 2: Do higher PageValues lead to a greater likelihood of purchase?

It is observed that users with higher PageValues are significantly more likely to generate revenue. This confirms that PageValues is a strong indicator of purchase intent and plays a crucial role in predicting user behavior.

Question 3: Do higher BounceRates negatively affect purchasing behavior?

The analysis shows that users with higher BounceRates tend to leave the website quickly without interacting further, which reduces the probability of making a purchase. Thus, higher BounceRates negatively impact conversions.

Question 4: Does spending more time on product-related pages increase the likelihood of purchase?

Users who spend more time on product-related pages are more engaged with the website and are more likely to make a purchase. This indicates that ProductRelated_Duration is an important feature influencing buying behavior.

Question 5: Does purchasing behavior vary across different months?

The analysis suggests that purchase activity varies across months, which may be influenced by seasonal trends, offers, or festive periods. Certain months show higher purchase rates compared to others.

Question 6: Does weekend activity influence purchasing behavior?

It is observed that user behavior differs slightly on weekends compared to weekdays. However, the impact of weekends on purchasing decisions is moderate and requires further analysis for stronger conclusions.

1.11 Implementation

The implementation of the machine learning models was carried out using Python and the Scikit-learn library. The dataset was first split into training and testing sets using an 80:20 ratio. The models were trained on the training data and evaluated on the testing

data. Various performance metrics such as accuracy, precision, recall, and F1-score were used to assess the effectiveness of the models.

The Random Forest classifier was used as the final model due to its superior performance. Feature importance was extracted from the trained model to identify the most influential variables affecting purchasing decisions.

1.12 Implementation Code

This section presents the implementation of the data preprocessing, exploratory data analysis, and machine learning models used in this project. The code is written in Python using libraries such as Pandas, NumPy, Matplotlib, Seaborn, and Scikit-learn.

1.12.1 Data Loading

```
import pandas as pd

df = pd.read_csv("online_shoppers.csv")
print(df.head())
print(df.info())
```

1.12.2 Data Preprocessing

```
# Handling missing values
df['BounceRates'].fillna(df['BounceRates'].mean(), inplace=True)
df['Month'].fillna(df['Month'].mode()[0], inplace=True)
df['PageValues'].fillna(df['PageValues'].median(), inplace=True)

# Encoding categorical variables
df = pd.get_dummies(df, drop_first=True)

# Splitting features and target
X = df.drop('Revenue', axis=1)
y = df['Revenue']
```

1.12.3 Train-Test Split

```
from sklearn.model_selection import train_test_split

X_train, X_test, y_train, y_test = train_test_split(
    X, y, test_size=0.2, random_state=42
```

```
)
```

1.12.4 Model Training

```
from sklearn.ensemble import RandomForestClassifier

model = RandomForestClassifier()
model.fit(X_train, y_train)

y_pred = model.predict(X_test)
```

1.12.5 Model Evaluation

```
from sklearn.metrics import accuracy_score, confusion_matrix,
    classification_report

print("Accuracy:", accuracy_score(y_test, y_pred))
print(confusion_matrix(y_test, y_pred))
print(classification_report(y_test, y_pred))
```

1.12.6 Feature Importance

```
import matplotlib.pyplot as plt

importances = model.feature_importances_
plt.barh(X.columns, importances)
plt.title("Feature Importance")
plt.show()
```

1.12.7 Visualization Code

```
import seaborn as sns
import matplotlib.pyplot as plt

# Heatmap
sns.heatmap(df.corr(), cmap='coolwarm')
plt.show()

# Revenue count
sns.countplot(x='Revenue', data=df)
```

```
plt.show()

# Page values distribution
sns.histplot(df['PageValues'], bins=50)
plt.show()
```

1.13 Visualizations

Various visualizations were created to better understand the dataset and support the analysis. These plots help in identifying patterns, relationships, and trends in user behavior.

As shown in Figure 1.2, the correlation heatmap illustrates the relationships between numerical features in the dataset. It shows that ProductRelated and ProductRelated_Duration have a strong positive correlation. Additionally, ExitRates and BounceRates are positively related. PageValues shows a strong association with the Revenue variable, indicating its importance in predicting purchases.

As observed in Figure 1.3, most users have low or zero PageValues, while only a few users have high values. This indicates that only a small portion of visitors contribute significantly to revenue, and higher PageValues are associated with a higher probability of purchase.

Figure 1.4 shows the distribution of users who made a purchase versus those who did not. It is evident that the majority of users did not make a purchase, indicating class imbalance in the dataset.

As illustrated in Figure 1.5, users with higher PageValues are more likely to generate revenue. This confirms that PageValues is one of the most significant predictors of purchasing behavior.

Figure 1.6 shows how purchasing behavior varies across different months. Certain months exhibit higher purchase activity, which may be influenced by seasonal trends or promotional events.

As seen in Figure 1.7, returning visitors are significantly more likely to make a purchase compared to new visitors. This highlights the importance of user retention in increasing conversions.

Figure 1.8 illustrates that users with higher bounce rates are less likely to make a purchase. This indicates that lower user engagement leads to reduced conversion rates.

As shown in Figure 1.9, the ROC curve demonstrates the performance of the classification model. The curve approaching the top-left corner indicates good model performance and strong discriminative ability.

Figure 1.10 compares the performance of different machine learning models. It is evident that the Random Forest model outperforms Logistic Regression and Decision Tree in terms of accuracy.

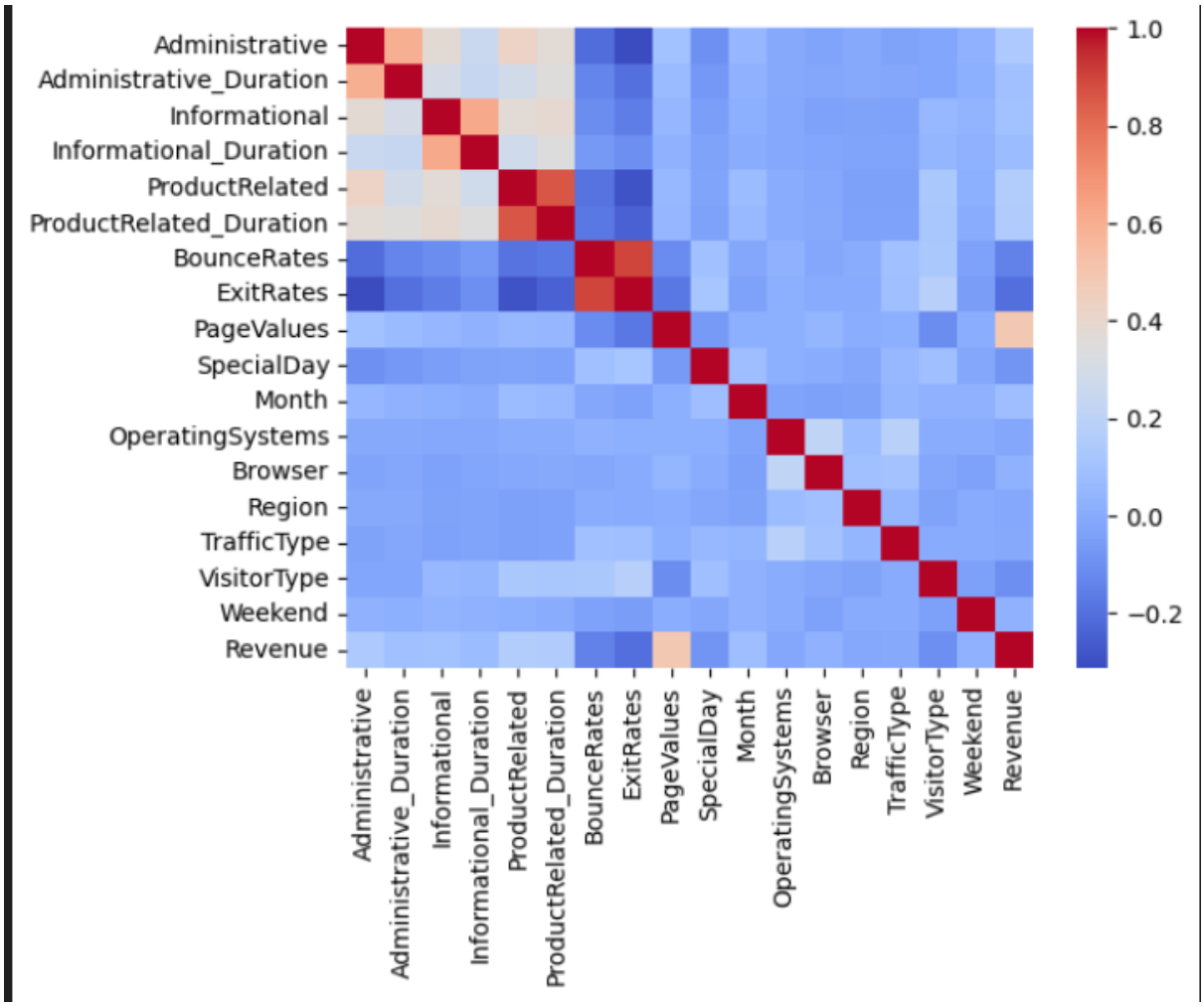

Figure 1.2: Correlation Heatmap

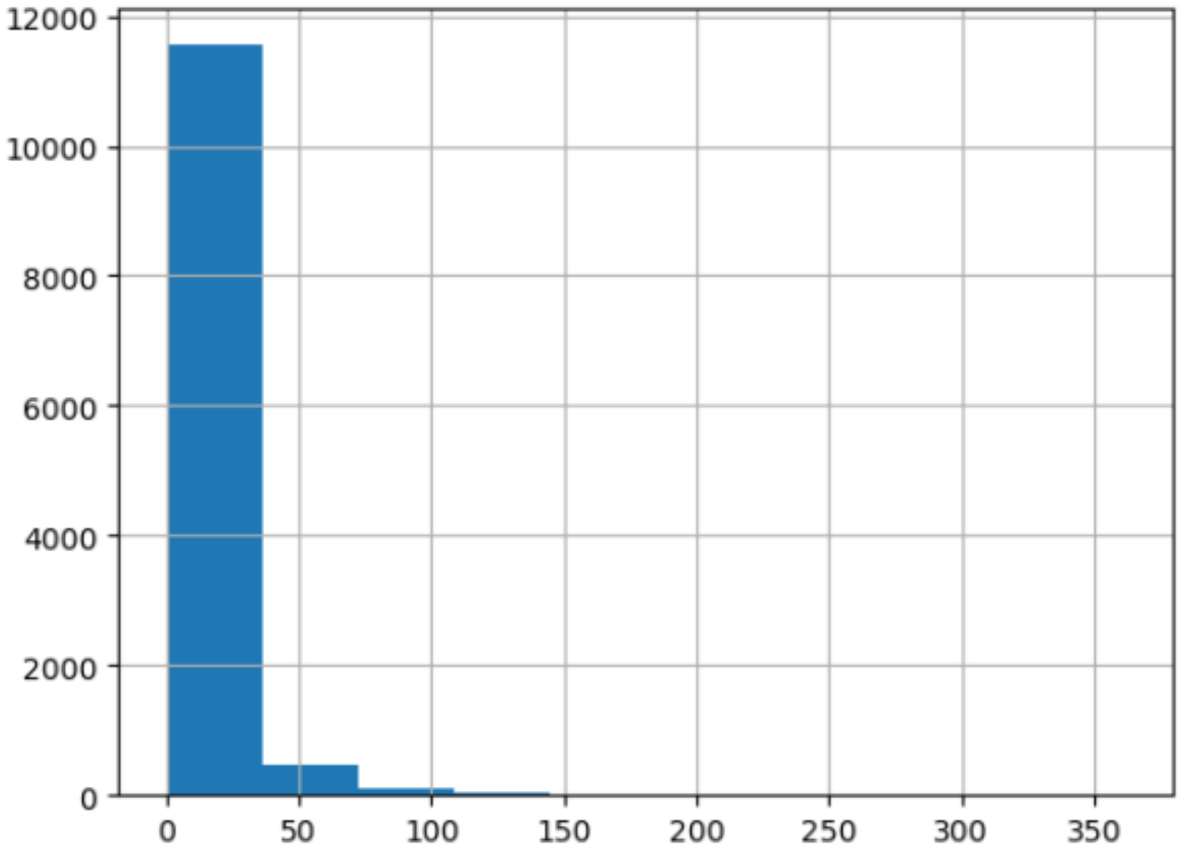

Figure 1.3: Distribution of Page Values

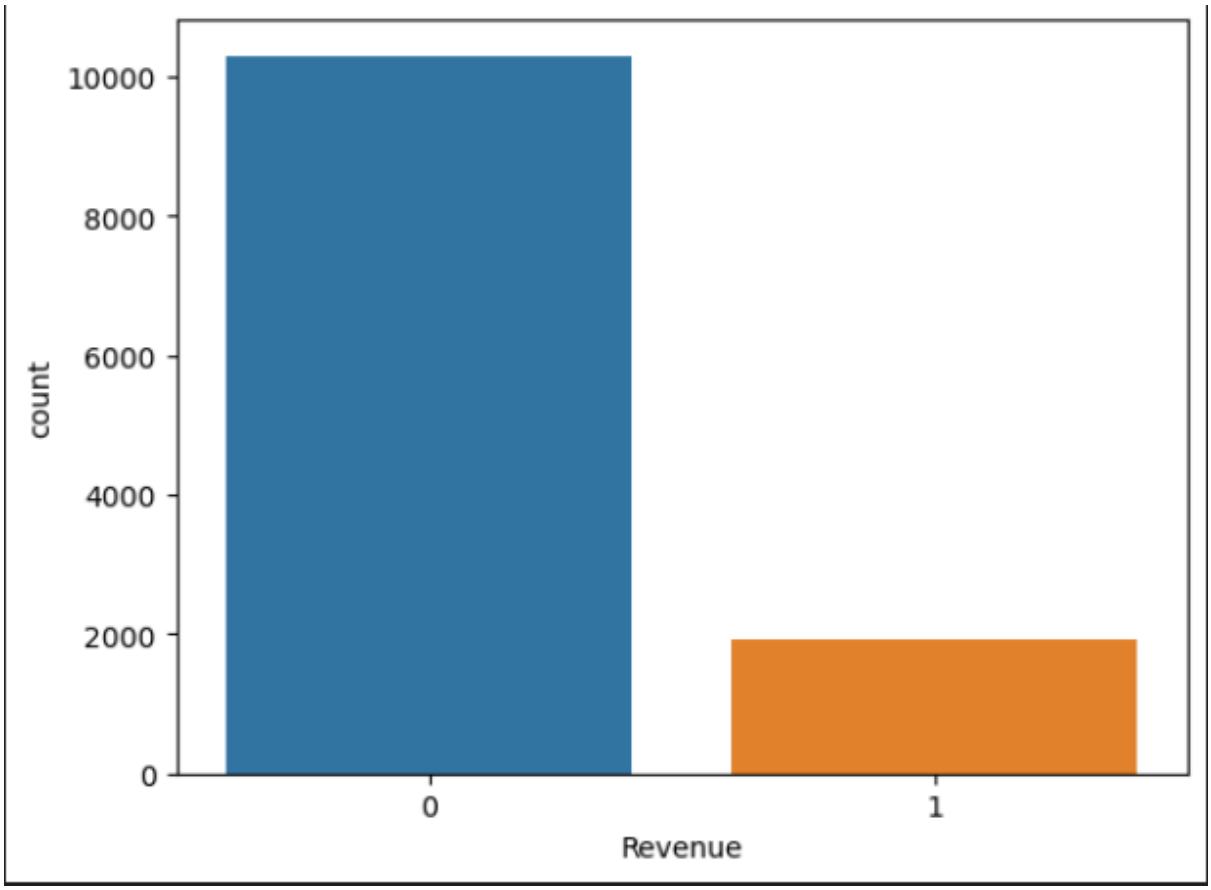

Figure 1.4: Revenue Count

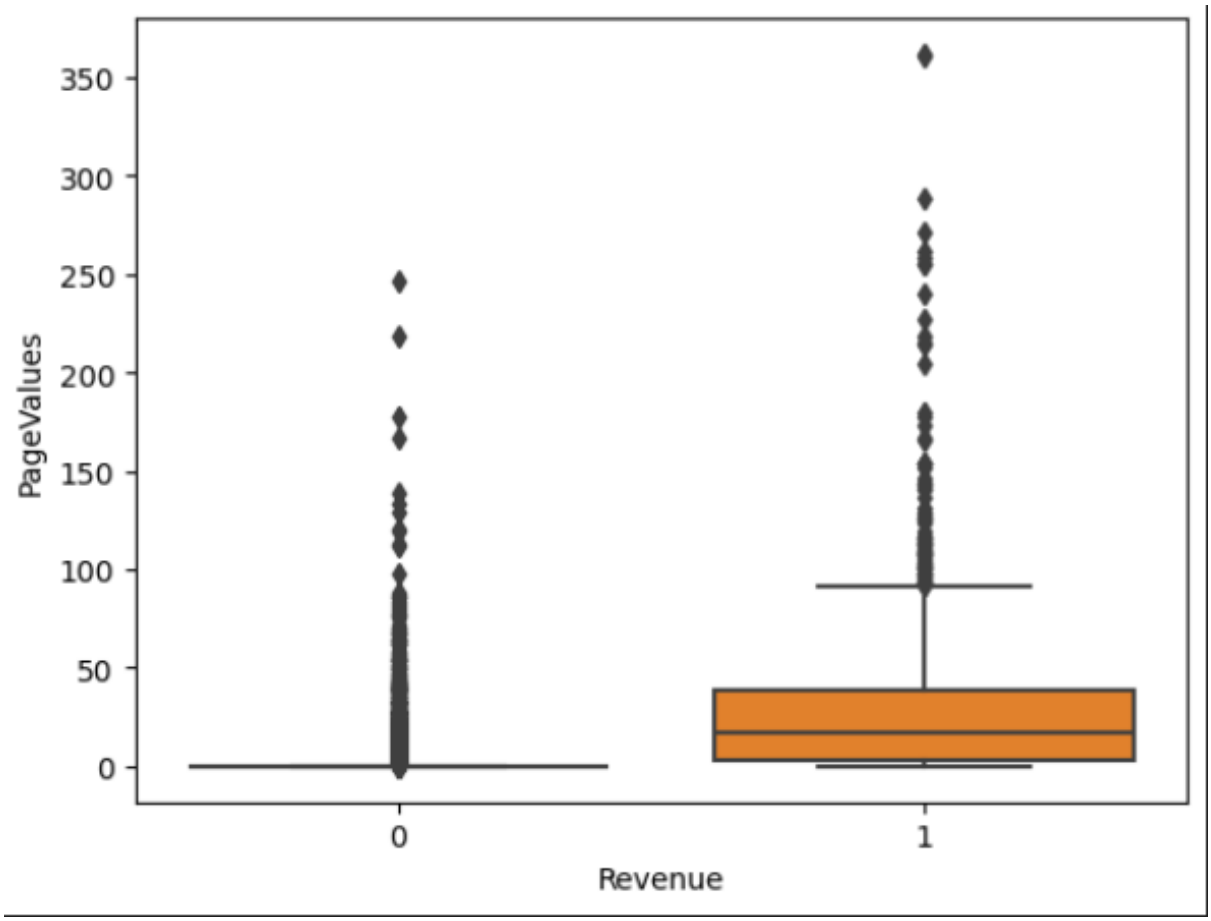

Figure 1.5: Revenue vs Page Values

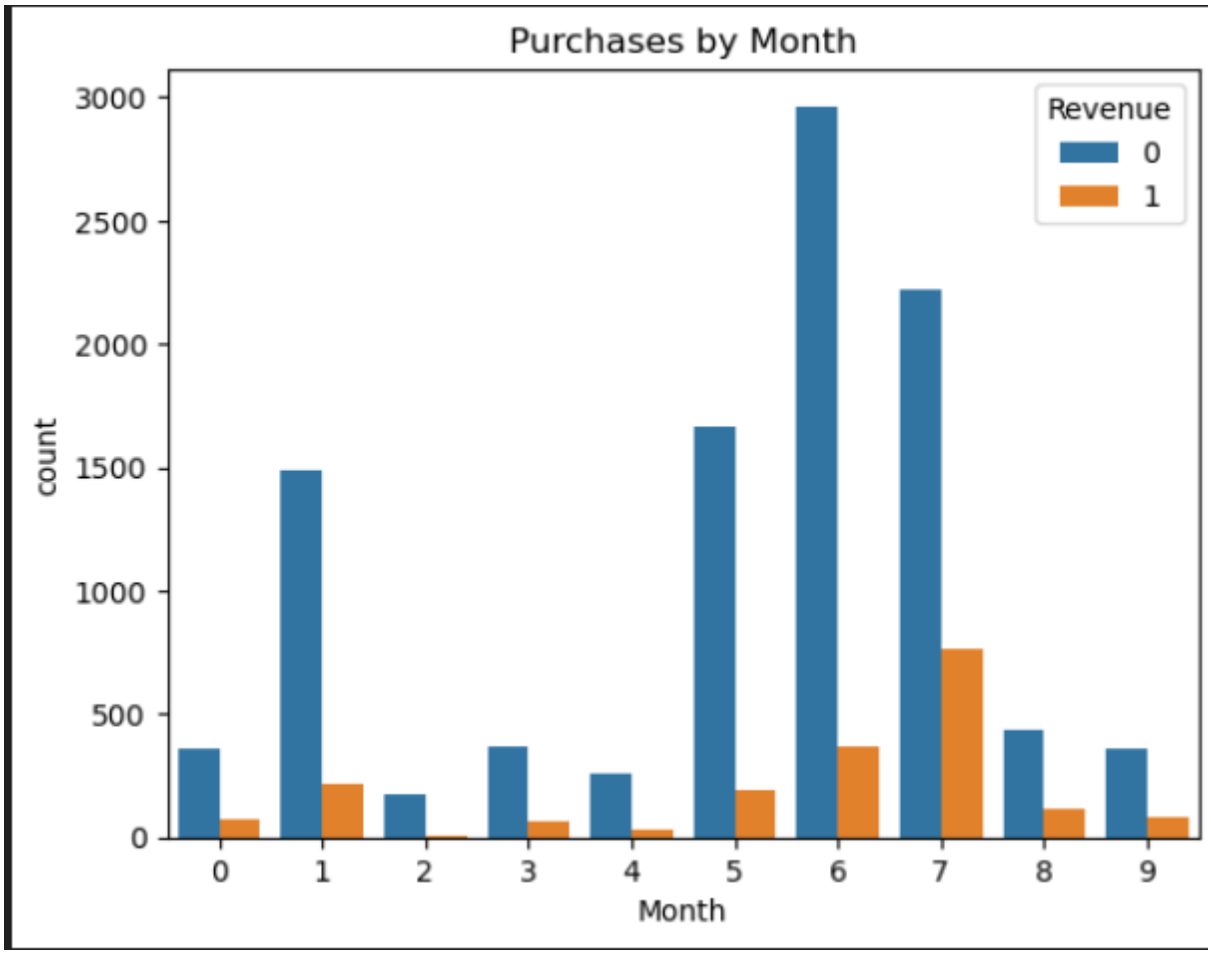

Figure 1.6: Purchase by Month

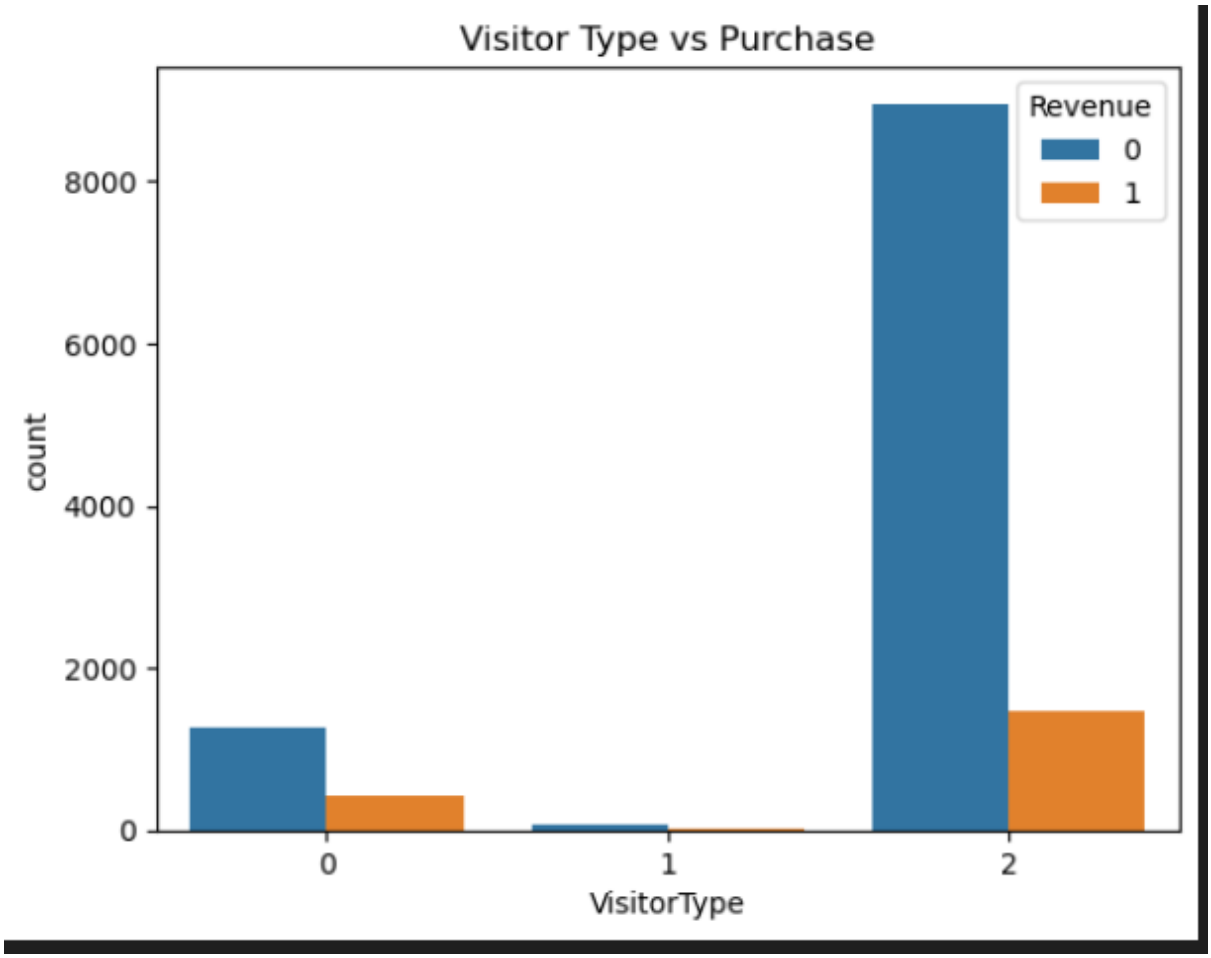

Figure 1.7: Visitor Type vs Purchase

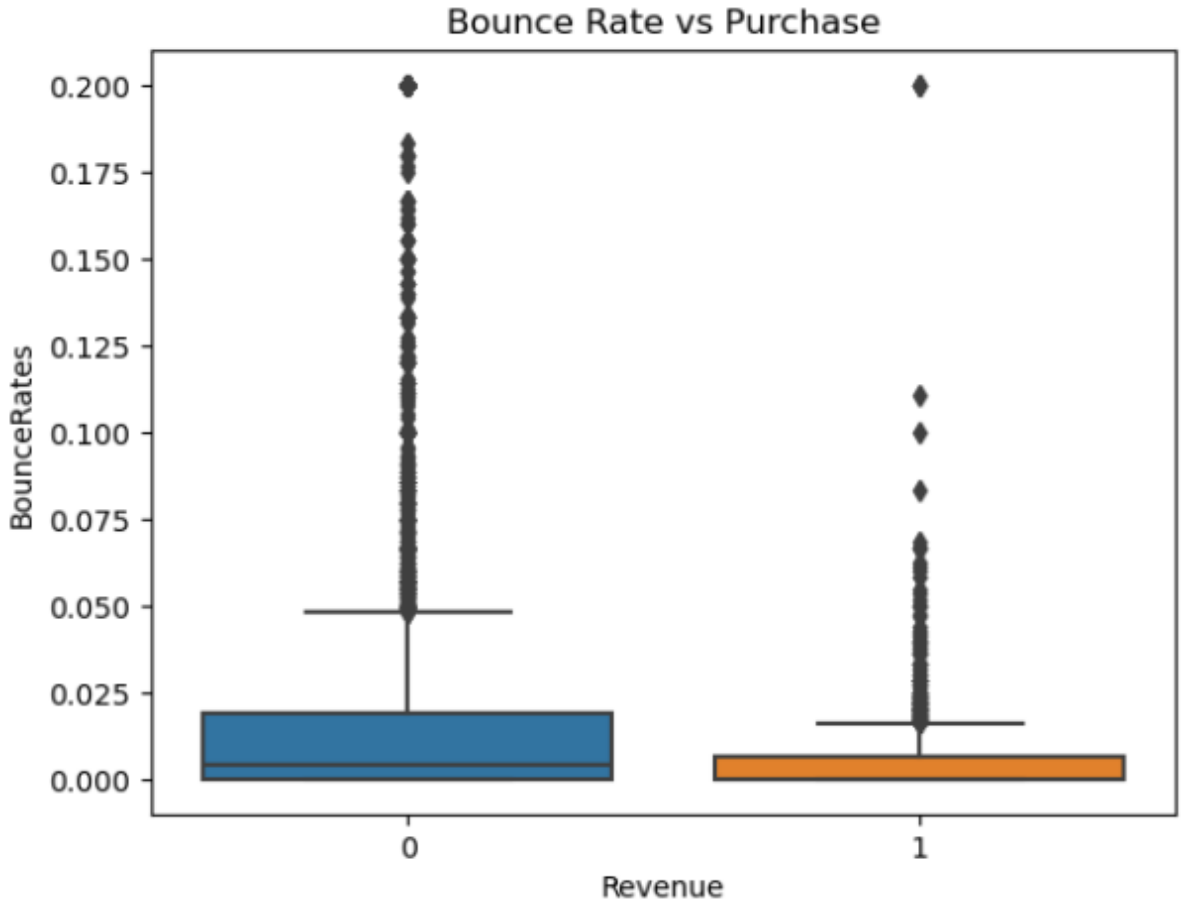

Figure 1.8: Bounce Rate vs Purchase

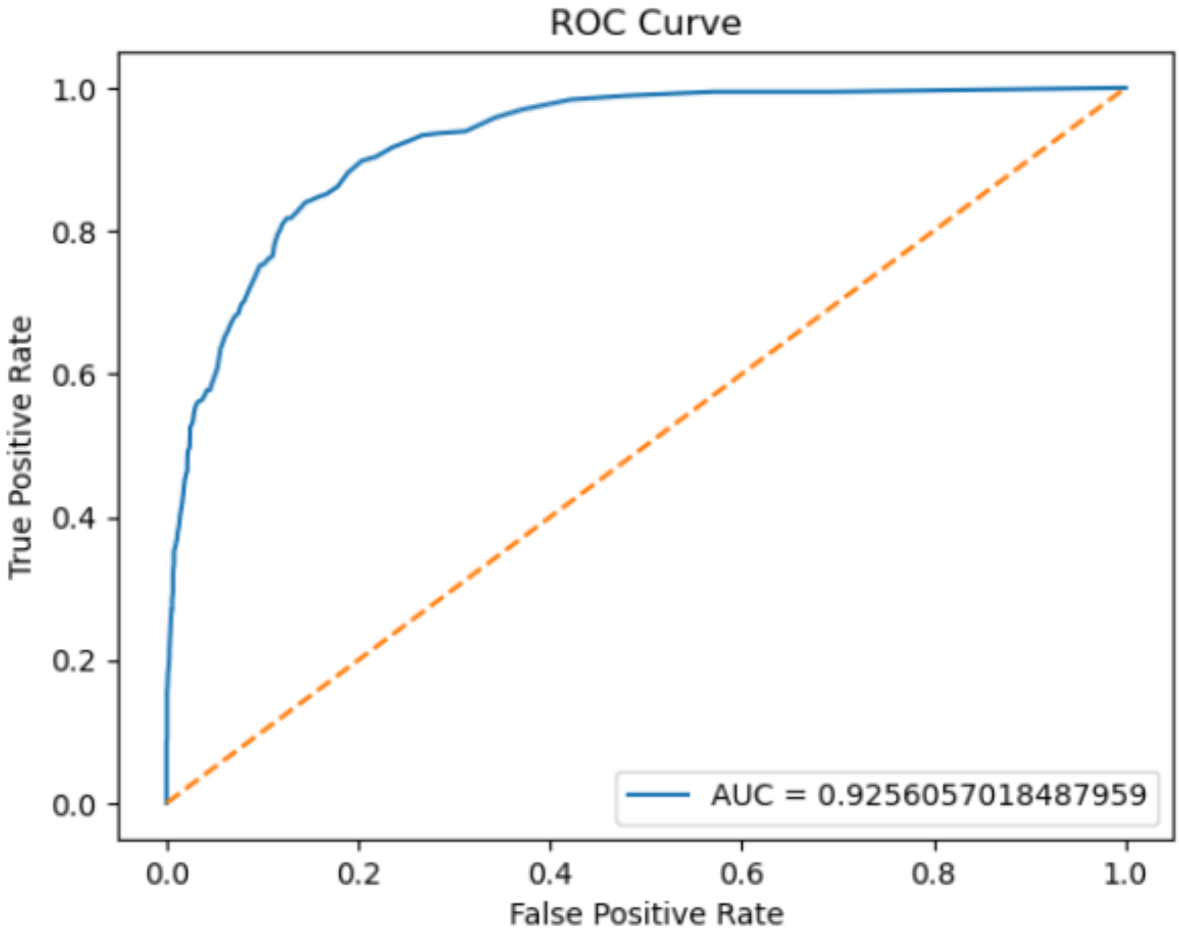

Figure 1.9: ROC Curve

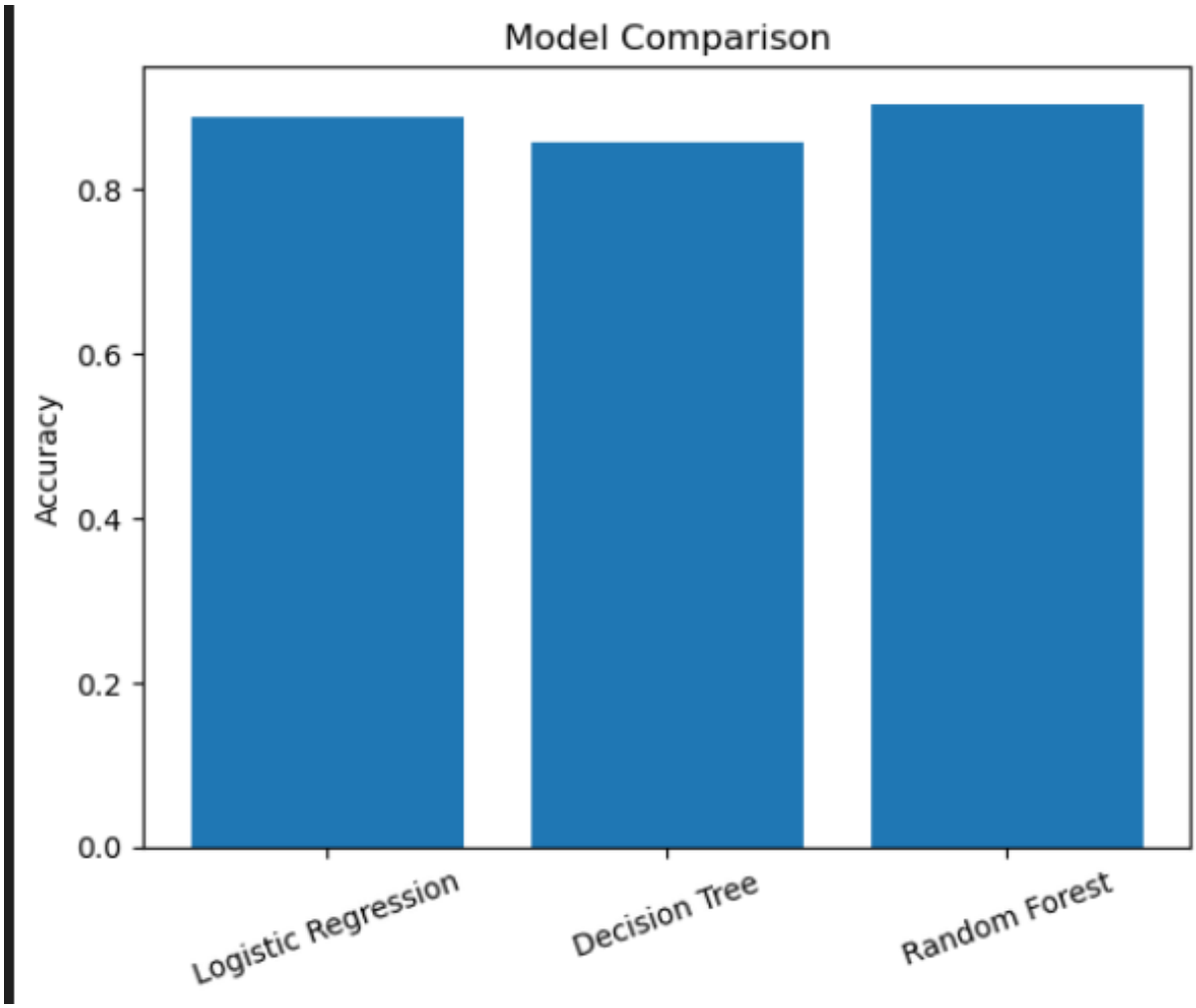

Figure 1.10: Model Comparison

The study concludes that Random Forest is the most effective model for predicting user purchasing behavior in this dataset. The analysis reveals that returning visitors and features such as PageValues have a strong influence on purchase decisions. The findings highlight the importance of user engagement metrics in understanding and predicting customer behavior.

1.14 Future Work

Future work can focus on improving the model performance by incorporating advanced techniques such as deep learning models. Including additional features such as user demographic information may further enhance prediction accuracy. Moreover, the development of real-time recommendation systems based on user behavior can provide personalized experiences and improve conversion rates.

1.15 References

This study is based on the Online Shoppers Purchasing Intention Dataset obtained from the UCI Machine Learning Repository. Additional references include the Scikit-learn documentation, which was used for implementing machine learning algorithms and evaluation metrics.

1.16 GitHub Repository

The complete implementation of this project is available at the following link:

`https://github.com/Tejurajan/DataScience_Project`